

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Machine Learning Aug 2025 to May 2026

- Boosted malware-family classification to **80.2%** accuracy across **17 families** by soft-voting **ResNet50**, **ViT-B/16**, VGG16, HOG, and Haralick models, beating the best single model by **2.4 pts** (McNemar $p < 0.001$).
- Generated **136,000** training images from **17,000** binaries across **8** visual representations and benchmarked **6** model families on **1,700** held-out samples.
- Proved a **2.4 pt** ensemble gain was real with **McNemar** testing at $p < 0.001$ instead of reporting a single accuracy number, across **1,700** held-out samples.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.

Turito | Software Engineering Intern, Systems Performance Jun 2022 to Oct 2022

- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Prototyped a **Rust + QUIC** media path serving about **8,000** lightweight subscribers with about **24%** lower startup time.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, SQL, Java

ML: PyTorch, ViT-B/16, ResNet50, VGG16, EfficientNet, ensembles, metric learning, scikit-learn, XGBoost, Hugging Face

LLM & Retrieval: RAG, embeddings, HNSW vector search, LLM serving gateways, OpenAI and Anthropic APIs

Infra: Docker, AWS, Kafka, Temporal, gRPC, PostgreSQL, Redis, Qdrant, Linux, GPU programming

PROJECTS

Contactless Palmprint Verification | *PyTorch, OpenCV, Siamese networks*

- Reached **0.9975 AUC** and **96.19% TAR at 1% FAR** by fusing 8 feature families into a 256-d triplet-loss embedding.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, GPU Programming, Biometric Security, Advanced Algorithms, Distributed Computing

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)